

Notes on Variance Calculations

The mathematical derivations showing how we can set achieve the following in our simulations:

- The variance of X , Y and C is 1.
- Each variable has a mean 0.
- The autoregressive effects of X_{t-1} on X_t is 0.25 (same for Y_{t-1} on Y_t).
- The lag-1 cross effect of X_{t-1} on Y_t is 0.10 (same for Y_{t-1} on X_t).
- There are 3 confounders, and hence $k = 3$.
- Their effects differ for X and Y , such that X is affected at $[0.30, 0.20, 0.10]$ and Y is affected at $[0.30, 0.20, 0.10]$.
- The innovations have residual correlations of 0.3 at each time-point, but not across time-points.
- At the first time-point ($T = 0$), X and Y are correlated due to the confounding variables.
- At the first time-point, we can assume that the dynamic process was going on before the data started being collected. Hence we want to simulate a system that is already stationary, but also where the first time-point has some 'extra' correlation of X and Y due to the fact that we assume the process was going before the start of the data collection.

Defining matrices

We have variables X over occasions T and Y over occasions T . This yields the vector:

$$Z_t = \begin{bmatrix} X_t \\ Y_t \end{bmatrix}$$

For one individual i , the vector looks like:

$$Z_i = \begin{bmatrix} X_{i1} \\ Y_{i1} \\ X_{i2} \\ Y_{i2} \\ X_{i3} \\ Y_{i3} \\ X_{i4} \\ Y_{i4} \\ X_{i5} \\ Y_{i5} \end{bmatrix}.$$

Expanding for N individuals:

$$Z = \begin{bmatrix} X_{1,1} & Y_{1,1} & X_{1,2} & Y_{1,2} & X_{1,3} & Y_{1,3} & X_{1,4} & Y_{1,4} & X_{1,5} & Y_{1,5} \\ X_{2,1} & Y_{2,1} & X_{2,2} & Y_{2,2} & X_{2,3} & Y_{2,3} & X_{2,4} & Y_{2,4} & X_{2,5} & Y_{2,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{N,1} & Y_{N,1} & X_{N,2} & Y_{N,2} & X_{N,3} & Y_{N,3} & X_{N,4} & Y_{N,4} & X_{N,5} & Y_{N,5} \end{bmatrix}.$$

Now for time-point t , X_t and Y_t are made up of some part of the confounder effect, and some part by the dynamic effects. With the dynamic effects here we mean every effect that 'feeds forward' to the next time-point. For the first time-point, there are no dynamic effects at all, since there are no previous time-points

. Another way of phrasing this is that some part of the variance in X (or Y) is explained by the confounders, some part

We can capture this in the formula:

$$Z_t = AZ_{t-1} + BU + E_t,$$

where

- A is the matrix of autoregressive and cross-lagged effects,
- B is the matrix of confounder effects,
- U is the vector of time-invariant confounders,
- E_t is the vector of innovations at time-point t .

If we choose our effects as denoted above we get the following A matrix:

$$A = \begin{bmatrix} a_x & b_y \\ b_x & a_y \end{bmatrix} = \begin{bmatrix} 0.25 & 0.10 \\ 0.10 & 0.25 \end{bmatrix}$$

We get the following beta matrix:

$$B = \begin{bmatrix} c_{x1} & c_{x2} & c_{x3} \\ c_{y1} & c_{y2} & c_{y3} \end{bmatrix} = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix}$$

And matrix U :

$$U = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} \\ c_{2,1} & c_{2,2} & c_{2,3} \\ c_{3,1} & c_{3,2} & c_{3,3} \\ \vdots & \vdots & \vdots \\ c_{N,1} & c_{N,2} & c_{N,3} \end{bmatrix}$$

Since the innovations matrix needs to be specified based on the other matrixes, we need to calculate the values that go into it to make the variances of each variable 1.

Calculating variances

As we just denoted, each measurement occasion takes some variance from the confounders and some from the dynamic effect. The variance received from the confounders is time-invariant, which means we can calculate it directly. Since the variances of all variables (and thus also the confounders) is 1, and the confounders are not correlated with each other, their covariance matrix is the identity matrix I . Hence, the variance received from the confounders is:

$$S_U = B\Psi B^T = BIB^T = BB^T$$

If we use the values from above, we get:

$$S_U = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix} \begin{bmatrix} 0.30 & 0.30 \\ 0.20 & 0.20 \\ 0.10 & 0.10 \end{bmatrix} = \begin{bmatrix} 0.14 & 0.14 \\ 0.14 & 0.14 \end{bmatrix}$$